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| **Started on** | Sunday, 24 July 2022, 4:49 PM |
| **State** | Finished |
| **Completed on** | Sunday, 24 July 2022, 4:50 PM |
| **Time taken** | 10 secs |
| **Marks** | 0.00/64.00 |
| **Grade** | **0.00** out of 100.00 |

Question **1**

Not answered

Marked out of 1.00

Flag question

The internal capsule is supplied by which of the following structures?

Select one:

- Anterior cerebral artery
- Basilar artery
- Posterior inferior cerebellar artery
- Middle cerebral artery
- Circle of Willis

The internal carotid artery enters the circle of Willis and divides to form the anterior cerebral and middle cerebral arteries. The anterior cerebral artery supplies the medical and superior lateral aspects of the cerebral cortex to the parietal/occipital border. The middle cerebral artery supplies the greater part of the lateral aspect of the cerebral cortex. The middle cerebral artery supplies the Broca's and Wernicke's area in the dominant hemispheres. The posterior cerebral artery supplies the inferomedial temporal lobe and the occipital lobe. Internal capsule contains traversing corticospinal tract and receives blood supply from different parts of the circle of Willis. The superior portion of the anterior limb is supplied by lenticulostriate branches of the middle cerebral artery while the inferior half is supplied by the recurrent artery of Heubner arising from the anterior cerebral artery. The genu receives supply from lenticulostriate branches of middle cerebral artery, while posterior limb is supplied by the lenticulostriate branches of middle cerebral artery (superior half) and anterior choroidal artery off of the internal carotid artery (inferior half).

The correct answer is: Circle of Willis

Question **2**

Not answered

Marked out of 1.00

Flag question

OCD is associated with an abnormality in which of the following parts of the brain?

Select one:

- Inferior olivary nucleus
- Medial temporal lobe
- Dendate nucleus
- Lentiform nucleus
- Red nucleus

Basal ganglia dysfunction is implicated in OCD. The components of basal ganglia are the caudate nucleus, the putamen and globus pallidus. Putamen and globus pallidus together form the lenticular/lentiform nuclei. Neuroimaging studies with PET scan suggest the involvement of both caudate and lentiform nuclei dysfunction in patients with OCD.

The correct answer is: Lentiform nucleus

Question **3**

Not answered

Marked out of 1.00

Flag question

Which area of the brain is responsible for normal arithmetical ability?

Select one:

- Fusiform gyrus
- Angular gyrus
- Cuneus gyrus
- Cingulate gyrus
- Supramarginal gyrus

Explanation:

The angular gyrus is situated in the parietal lobe, that lies near the upper boundary of the temporal lobe, and immediately posterior to the supramarginal gyrus. 'Acalculia' refers to impaired arithmetic skills. Gerstmann syndrome consists of four neuropsychological disturbances: acalculia, finger agnosia, left/right confusion, and agraphia. When all four elements occur, neurologists usually attribute the syndrome to a stroke or other lesion in the angular gyrus of the dominant parietal lobe.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 167.

The correct answer is: Angular gyrus

Question **4**

Not answered

Marked out of 1.00

Flag question

Which one of the following types of neurons has no axons?

Select one:

- Golgi type 1 neurons
- Amacrine neurons
- Polysynaptic neurons
- Golgi type 2 neurons
- Bipolar neurons

Neuronal cells are classified on the basis of their size into three types: Golgi type 1 - Long axon; Golgi type 2 - Short axon terminating near the parent cell; Amacrine neurons - No axon.

The correct answer is: Amacrine neurons

Question **5**

Not answered

Marked out of 1.00

Flag question

Which of the following statements regarding amygdala is NOT correct?

Select one:

- Damage leads to impaired fear conditioning
- Emotional face processing involves amygdalar activity
- It is involved in emotional memory
- Learning does not occur in the absence of memory consolidation mediated by amygdala
- Monkeys with amygdalar damage show impaired maternal behaviours

Amygdala is involved in rating the emotional importance of an experience and activating the level of hippocampal response accordingly. Amygdalar damage leads to loss of fear conditioning and in monkeys, loss of maternal behaviour has also been noted. Despite the amygdalar damage, learning and consolidation of memory can occur, especially in the absence of emotional valence and arousal.

The correct answer is: Learning does not occur in the absence of memory consolidation mediated by amygdala

Question **6**

Not answered

Marked out of 1.00

Flag question

The limbic Papez circuit includes which of the following structures?

Select one:

- Hippocampus
- Temporal pole
- Medullary centre
- Trapezoid body
- Red nucleus

Explanation:

The limbic system was delineated by James Papez in 1937. The Papez circuit consists of the hippocampus, the fornix, the mammillary bodies, the anterior nucleus of the thalamus, the cingulate gyrus, and the parahippocampal gyrus. The boundaries of the limbic system were subsequently expanded to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 934.

The correct answer is: Hippocampus

Question **7**

Not answered

Marked out of

Which one of the following effects is associated with lesions of the left hemisphere?

Select one:

1.00

Flag question

Prosopagnosia

- Apraxia
- Autotopagnosia
- Alexia with agraphia
- Visuospatial agnosia

Explanation:

Alexia is an inability to read written or printed words, usually called ‘dyslexia’. Strokes, trauma, or other lesions of the dominant (usually left) hemisphere may impair reading ability (neurologists call this condition ‘acquired dyslexia’). Right-sided motor deficits usually accompany acquired dyslexia (alexia), as does agraphia (inability to write).

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 22.

Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman’s Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 166.

The correct answer is: Alexia with agraphia

Question **8**

Not answered

Marked out of 1.00

Flag question

Which of the following is correct with respect to cerebral dominance for language?

Select one:

- Most left handed individuals have bilateral dominance
- Most left handed individuals have right cerebral dominance
- Most left handed individuals have left cerebral dominance
- Right handed individuals always have right cerebral dominance
- Dominance cannot be ascertained in left handed individuals

Explanation:

In most people, the hemisphere dominant for language also directs the dominant hand. Ninety percent of the population is right-handed, and 99% of right-handers have left hemispheric dominance for language. Of the 10% who are left-handers, 67% also have left-hemispheric language dominance; the other 33% have either mixed or right-hemispheric language dominance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 931.

The correct answer is: Most left handed individuals have left cerebral dominance

Question **9**

Not answered

Marked out of 1.00

Flag question

Which one of the following cells is considered as descendants of macrophages that clear neuronal debris following cell death?

Select one:

- Microglial cells
- Ependymal cells

- Schwann cells
- Astrocytes
- Oligodendrocytes

Explanation:

There are three types of glial cells. The most common are the **astrocytes**, which have several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. The **oligodendrocytes** in the central nervous system and the **Schwann cells** in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia,** which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Microglial cells

Question **10**

Not answered

Marked out of 1.00

Flag question

Which structure separates the two lateral ventricles in the human brain?

Select one:

- Septum pellucidum
- Pons
- Aqueduct of sylvius
- Septum pallidum
- Corpus callosum

The body of the lateral ventricle lies immediately below the corpus callosum and they are separated by septum pellucidum. An anomalous splitting of the septum pellucidum may be more common in schizophrenia than in general population (cavum septum pellucidum). The third ventricle lies between thalamus and hypothalamus. The fourth ventricle lies above the pons and just below the cerebellum. The Aqueduct of Sylvius links the third and fourth ventricles.

The correct answer is: Septum pellucidum

Question **11**

Not answered

Which of the following cells are uniquely found in the cerebellum?

Select one:

Marked out of 1.00

Flag question

- Purkinje cells
- Betz cells
- Pyramidal cells
- Granule cells
- Stellate cells

Purkinje cells are a class of GABAergic neurons located in the cerebellar cortex only. Purkinje cells send inhibitory projections to the deep cerebellar nuclei and constitute the sole output of all motor coordination in the cerebellar cortex.

The correct answer is: Purkinje cells

Question **12**

Not answered

Marked out of 1.00

Flag question

The foramen of Monro connects

Select one:

- Fourth ventricle to subarachnoid space
- Lateral ventricle to third ventricle
- Fourth ventricle to central canal of spinal cord
- Lateral ventricle to fourth ventricle
- Lateral ventricle to second ventricle

The foramen of Monro (also known as the interventricular foramen) is part of the ventricular system and the connection between the third ventricle and the lateral ventricles. It allows for the flow of CSF, and, therefore, if blocked or effaced, causes a reduction in CSF flow from the lateral ventricles despite no associated reduction in CSF production from the choroid plexus.

The correct answer is: Lateral ventricle to third ventricle

Question **13**

Not answered

Marked out of 1.00

Flag question

Which part of the brain is last to mature?

Select one:

- Amygdala
- Prefrontal cortex
- Midbrain
- Parahippocampal cortex
- Thalamus

Explanation:

One of the last places in the brain to mature, the prefrontal cortex is thought of as the

'personality centre' and is the cortical region that makes us uniquely human. Lesions in this area produce several neuropsychiatric disorders that tend to present with disinhibition, apathy, loss of initiative, and personality changes.

The correct answer is: Prefrontal cortex

Question **14**

Not answered

Marked out of 1.00

Flag question

Which artery supplies most of the corpus callosum?

Select one:

- Middle cerebral artery
- Anterior cerebral artery
- Posterior inferior cerebral artery
- Posterior cerebral artery
- Basilar artery

Explanation:

Infarcts of the corpus callosum are not common due likely to a rich blood supply from the main arterial systems, specifically the anterior cerebral and anterior communicating, but also from posterior cerebral arteries. The 'anterior cerebral artery syndrome', a disconnection syndrome, results from an occlusion of both anterior cerebral arteries that causes an infarction of both frontal lobes and the anterior corpus callosum (the genu).

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 172.

The correct answer is: Anterior cerebral artery

Question **15**

Not answered

Marked out of 1.00

Flag question

Regarding the blood supply of language areas which of the following is correct?

Select one:

- Wernicke area is supplied by the posterior cerebral artery
- Transient ischemia of the carotid system produces aphasia
- Transient ischemia of the carotid system produces dysarthria
- Vertebrobasilar system supplies the motor speech area
- Right-sided carotid block produces Broca aphasia

Explanation:

The brain receives blood from two sources: the internal carotid arteries and the vertebral arteries. The internal carotid arteries branch to form two major cerebral arteries, the anterior and middle cerebral arteries, which form the anterior circulation that supplies the forebrain. Brief loss of blood supply (ischaemia) can cause cellular changes which, if not quickly reversed, can lead to cell death. Historically, studies of the functional consequences of strokes, and their relation to vascular territories in the brain, provided information about the location of various brain functions. The location of major language centres in the left hemisphere, for instance, was discovered in this way. The carotid system is responsible for

supplying language areas. Ischaemia can produce aphasia.

Ref: Purves D, Augustine GJ, Fitzpatrick D, et al., editors. Neuroscience. 2nd edition. Sunderland (MA): Sinauer Associates; 2001. The Blood Supply of the Brain and Spinal Cord. Available from: https://www.ncbi.nlm.nih.gov/books/NBK11042/

The correct answer is: Transient ischemia of the carotid system produces aphasia

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following structures divides the frontal lobe from the temporal lobe?

Select one:

- Central sulcus
- Lateral sulcus
- Cingulate gyrus
- Calcarine fissure
- Superior temporal gyrus

The lateral sulcus (also called Sylvian fissure or lateral fissure) is a prominent landmark of the human brain. It divides the frontal lobe and parietal lobe from the temporal lobe.

The correct answer is: Lateral sulcus

Question **17**

Not answered

Marked out of 1.00

Flag question

In which area of the brain is the dentate gyrus located?

Select one:

- Hippocampus
- Hypothalamus
- Dorsolateral prefrontal cortex
- Amygdala
- Red nucleus

Explanation:

The hippocampus mediates several higher brain functions, such as learning, memory, and spatial coding. The input region of the hippocampus, the dentate gyrus, plays a critical role in these processes. Several lines of evidence suggest that the dentate gyrus acts as a preprocessor of incoming information, preparing it for subsequent processing. The dentate gyrus is one of the few brain regions exhibiting adult neurogenesis. Thus, new neurons are generated and functionally integrated throughout life.

Ref: Jonas P, Lisman J. Structure, function, and plasticity of hippocampal dentate gyrus microcircuits. Frontiers in neural circuits. 2014 Sep 10;8:107.

The correct answer is: Hippocampus

Question

**18**

Not answered

Marked out of 1.00

Flag question

Cranial nerves can carry both sensory and motor impulses. Which one of the following cranial nerves has only a motor component?

Select one:

- Vestibulocochlear
- Glossopharyngeal
- Olfactory
- Optic
- Abducens

Explanation:

The abducens nerves (sixth cranial nerves) originate in the pons. They perform only a single function and innervate only a single muscle. Each abducens nerve innervates its ipsilateral lateral rectus muscle, which abducts that eye. Abducens nerve impairment, which is relatively common, leads to inward deviation (adduction) of the eye, due to the unopposed medial pull of the medial rectus, but does not cause ptosis or pupil changes.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 28.

The correct answer is: Abducens

Question **19**

Not answered

Marked out of 1.00

Flag question

Forced utilization behaviour can be a feature of damage to which of the following structures?

Select one:

- Cerebellar damage
- Parietal lobe damage
- Temporal lobe damage
- Occipital lobe damage
- Frontal lobe damage

If the frontal lobe superior to the eye (orbitofrontal cortex) is damaged, forced utilization can be seen. When objects are placed in front of a subject with frontal damage, the subject will pick them and start using them even when the subject is instructed not to use them.

The correct answer is: Frontal lobe damage

Question **20**

Not answered

Marked out of 1.00

Flag question

Which lobe is initially affected in Alzheimer disease?

Select one:

- Prefrontal lobe
- Temporal lobe
- Frontal lobe

- Parietal lobe
- Occipital lobe

Explanation:

The entorhinal cortex, an essential component of the medial temporal lobe long-term-memory system, represents the main source of input to the hippocampus and the primary target of hippocampal outputs. The involvement of the entorhinal cortex is relevant for neurodegenerative disorders such as Alzheimer's disease (AD), as it is one of the earliest affected brain regions. The typical hallmarks of AD, such as the presence of amyloid protein and neurofibrillary tangles, are seen primarily in the entorhinal cortex in mild AD and 'spread' to the hippocampus and other cortical areas as the disease progresses.

Ref: Criscuolo C et al. Entorhinal Cortex dysfunction can be rescued by inhibition of microglial RAGE in an Alzheimer's disease mouse model. Scientific reports. 2017 Feb 13;7(1):1-5.

The correct answer is: Temporal lobe

Question **21**

Not answered

Marked out of 1.00

Flag question

The most common type of neuronal cells seen in the cerebral cortex are

Select one:

- Stellate cells
- Spindle cells
- Granule cells
- Purkinje cells
- Pyramidal cells

The pyramidal neurons with their triangular-shaped cell bodies make up nearly 75% of the cortical neurons. Stellate cells (25%) are present in all the layers except layer 1.

The correct answer is: Pyramidal cells

Question **22**

Not answered

Marked out of 1.00

Flag question

Which part of the brain is associated with aura in epilepsy?

Select one:

- Frontal lobe
- Pre-central gyrus
- Occipital lobe
- Temporal lobe
- Parietal lobe

Explanation:

The early warning or first phase of a seizure may be an aura (Latin, breeze). These are due to the onset of the seizure in a circumscribed part of the temporal lobe. Olfactory

hallucinations, for example, may represent the first phase or aura of seizures that originate in the medial-inferior surface of the temporal lobe, the uncus. These auras usually consist of a series of several-second episodes of ill-defined and unpleasant, but occasionally sweet or otherwise pleasant, smells preceding or superimposed on impaired consciousness and behavioural disturbance.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 26.

The correct answer is: Temporal lobe

Question **23**

Not answered

Marked out of 1.00

Flag question

Which of the following nuclei of basal ganglia contains melanin pigment?

Select one:

- Globus pallidus
- Substantia nigra
- Caudate nucleus
- Subthalamic nuclei
- Putamen

Explanation:

The substantia nigra is named the black substance because the presence of melanin pigment causes it to appear black to the naked eye. It has two parts, one of which is functionally equivalent to the globus pallidus interna. The other part degenerates in Parkinson disease. Depigmentation of the substantia nigra is seen in Parkinson disease.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 929.

The correct answer is: Substantia nigra

Question **24**

Not answered

Marked out of 1.00

Flag question

Nominal aphasia can be localised to

Select one:

- Angular gyrus
- Posterior parietal cortex
- Inferior frontal gyrus
- Superior temporal gyrus
- Broca area

Explanation:

In nominal aphasia the patient cannot name objects, although they have plenty of words at their disposal. Usually, they find it difficult to carry out verbal and written commands and they cannot write spontaneously, although as a rule they can copy written material. Difficulty in finding the correct word can occur in other varieties of aphasia, but in nominal aphasia it is the outstanding disorder. Nominal aphasia may be found with either diffuse brain damage or

with focal lesions involving the dominant temporoparietal region (e.g. angular gyrus).

Ref: Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 59.

The correct answer is: Angular gyrus

Question **25**

Not answered

Marked out of 1.00

Flag question

Which function of the brain is served by angular gyrus?

Select one:

- Balance & co-ordination
- Motor movements
- Language
- Memory
- Executive functions

The angular gyrus is a region of the inferior parietal lobe of the brain that is involved in the processing of auditory and visual input and in the comprehension of language. It is Brodmann area 39 of the human brain. Lesions are associated with anomia.

The correct answer is: Language

Question **26**

Not answered

Marked out of 1.00

Flag question

Primary sensory cortex is located in which of the following areas?

Select one:

- Superior temporal area
- Inferior parietal lobule
- Dorsolateral prefrontal cortex
- Precentral gyrus
- Postcentral gyrus

Brodmann areas 3, 1 and 2 comprise the primary somatosensory cortex of the human brain (or S1) - most of which is located on the postcentral gyrus. Lesions affecting the primary somatosensory cortex produce characteristic symptoms including: agraphesthesia, astereognosia, loss of vibration, proprioception and fine touch. The primary motor cortex (or M1) is a brain region that in humans is located in the posterior portion of the frontal lobe. It works in association with pre-motor areas to plan and execute movements. M1 is located on precentral gyrus.

The correct answer is: Postcentral gyrus

Question **27**

Not answered

Marked out of

Which of the following cells are the largest of all neurons in the human brain?

Select one:

1.00

Flag question

- Betz cells
- Basket cells
- Golgi cells
- Purkinje cells
- Stellate cells

Large pyramidal cells called Betz cells are seen in the primary motor cortex.Betz cells are pyramidal cell neurons located within the fifth layer of the grey matter in the primary motor cortex. These can reach a diameter of 100 μm in some cases, and are known to be the largest neurons in the nervous system. Betz cells represent about 10% of the total pyramidal cell population in layer V of the human primary motor cortex.

The correct answer is: Betz cells

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a part of Papez circuit?

Select one:

- Mammillary body
- Parahippocampal gyrus
- Cingulate gyrus
- Hippocampus
- Heschl's gyrus

Explanation:

The limbic system was delineated by James Papez in 1937. The Papez circuit consists of the hippocampus, the fornix, the mammillary bodies, the anterior nucleus of the thalamus, the cingulate gyrus, and the parahippocampal gyrus. The boundaries of the limbic system were subsequently expanded to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 934.

The correct answer is: Heschl's gyrus

Question **29**

Not answered

Marked out of 1.00

Flag question

Which one of the following is seen as a key function of the non-dominant cerebral hemisphere?

Select one:

- Logical reasoning
- Pictorial memory
- Analytical capacity
- Verbal learning
- Language functions

Disorders of verbal and pictorial memory are commonly dissociated by brain injury with verbal memory being affected in left and picture memory predominantly by right brain injury; However, this is disputed as many subjects use verbal encoding when remembering pictures.

The correct answer is: Pictorial memory

Question **30**

Not answered

Marked out of 1.00

Flag question

A lesion affecting which of the following structures can cause dysarthria?

Select one:

- Homunculus
- Parietal lobe
- Temporal lobe
- Cerebellum
- Spinal cord

Explanation:

Extensive damage to the cerebellum causes ‘scanning speech’. A variety of dysarthria. Scanning speech, which reflects incoordination of speech production, is characterised by poor modulation, irregular cadence, and inability to separate adjacent sounds.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman’s Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 12.

The correct answer is: Cerebellum

Question **31**

Not answered

Marked out of 1.00

Flag question

Which area of the brain is associated with 'social valuation'?

Select one:

- Calcarine fissure
- Lingual gyrus
- Orbitofrontal cortex
- Anterior cingulate gyrus
- Hippocampal gyrus

Orbitofrontal cortex is often included as the brain region with a prominent role in social valuation, though many other regions such as the temporoparietal junction, anterior cingulate cortex and fusiform gyrus also play a part in this function. Damage to OFC can cause a pseudopsychopathic syndrome.

The correct answer is: Orbitofrontal cortex

Question **32**

Not answered

Marked out of 1.00

Flag question

Which one of the following structures is involved in the visual pathway?

Select one:

- Medial geniculate body
- Heschl's gyrus
- Inferior colliculus
- Lateral geniculate body
- Tegmentum

Explanation:

The optic radiations sweep from the lateral geniculate body to the occipital lobe's visual cortex. Some of the fibres sweep anteriorly from the lateral geniculate body into the anterior temporal lobe before heading posteriorly; this detour is known as Meyer's loop. These fibres convey visual information from the contralateral superior quadrant.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 280.

The correct answer is: Lateral geniculate body

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following statements about microglia is correct?

Select one:

- It is mainly responsible for myelination
- It constitutes 70% of glial population in brain
- It is mesodermal in origin
- It is sensitive to sodium channel stimulation
- It is present outside brain and spinal cord

Explanation:

Despite some controversies on microglia origin from either the mesoderm or ectoderm, several investigators have presented evidence supporting a mesodermal origin of microglia. Because of their mesodermal origin, microglia share many features with other myeloid cells in the body.

Ref: Ginhoux F, Prinz M. Origin of microglia: current concepts and past controversies. Cold Spring Harbor perspectives in biology. 2015 Aug 1;7(8):a020537.

The correct answer is: It is mesodermal in origin

Question **34**

Not answered

Marked out of 1.00

A 62-year-old gentleman presented to the outpatient department with a history of changes in personality, socially disinhibited behaviour, poor judgement and irritability. His carer also reported a recent history of urinary incontinence. Which part of the brain is most likely to be affected?

Flag question

Select one:

- Occipital lobe
- Frontal lobe
- Cerebellum
- Temporal lobe
- Parietal lobe

Explanation:

Frontal lobe dysfunction may consist of slowed thinking, poor judgement, decreased curiosity, social withdrawal, and irritability. Patients typically display apathetic indifference to experience that can suddenly explode into impulsive disinhibition.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 935.

The correct answer is: Frontal lobe

Question **35**

Not answered

Marked out of 1.00

Flag question

Which one of the following is not a function of astrocytes?

Select one:

- Maintaining the blood brain barrier
- Formation of the CNS myelin sheath
- Nutrition of neurons
- Breakdown of some neurotransmitters
- None of the listed options

Explanation:

There are three types of glial cells. The most common are the **astrocytes,** which have several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. The **oligodendrocytes** in the central nervous system and the **Schwann cells** in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia**, which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Formation of the CNS myelin sheath

Question **36**

Not answered

Marked out of 1.00

Flag question

Which of the following structures give rise to cerebral hemispheres during embryogenesis?

Select one:

- Myelencephalon
- Diencephalon
- Mesencephalon
- Metencephalon
- Telencephalon

Explanation:

At 4 weeks, the human brain is divided longitudinally into the prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). These three subdivisions or 'vesicles' divide further into five divisions by 5 weeks, consisting of the prosencephalon, which forms the telencephalon (including cortex, hippocampus, and basal ganglia) and diencephalon (thalamus, subthalamus, hypothalamus), the mesencephalon (midbrain, including the red nucleus), and the rhombencephalon, yielding metencephalon (pons and cerebellum) and myelencephalon (medulla).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 939.

The correct answer is: Telencephalon

Question **37**

Not answered

Marked out of 1.00

Flag question

Neural crest cells originate from

Select one:

- Endoderm
- Pharyngeal pouch
- Mesoderm
- Ectoderm
- Neural tube

Explanation:

Although derived from the ectoderm, the neural crest has sometimes been called the fourth germ layer because of its importance. The neural crest cells originate in the dorsal-most region of the neural tube. The neural crest cells migrate extensively to generate a prodigious number of differentiated cell types. These cell types include:

- Neurons and glial cells of the sensory, sympathetic, and parasympathetic nervous system
- Epinephrine-producing (medulla) cells of the adrenal gland
- Pigment-containing cells of the epidermis
- Many of the skeletal and connective tissue components of the head

Ref: http://www.ncbi.nlm.nih.gov/books/NBK10065/

The correct answer is: Neural tube

Question **38**

Not answered

Marked out of 1.00

Flag question

Which one among the following types of neuroglia is present in the peripheral nervous system?

Select one:

- Schwann cells
- Astrocytes
- Microglia
- Ependyma
- Oligodendrocytes

Explanation:

There are three types of glial cells. The most common are the **astrocytes**, which have several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. The **oligodendrocytes** in the central nervous system and the **Schwann cells** in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia,** which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Schwann cells

Question **39**

Not answered

Marked out of 1.00

Flag question

The thalamus and hypothalamus are constituents of which of the following embryological structures?

Select one:

- Myelencephalon
- Metencephalon
- Telencephalon
- Mesencephalon
- Diencephalon

Explanation:

At 4 weeks, the human brain is divided longitudinally into the prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). These three subdivisions or 'vesicles' divide further into five divisions by 5 weeks, consisting of the prosencephalon, which forms the telencephalon (including cortex, hippocampus, and basal ganglia) and diencephalon (thalamus, subthalamus, hypothalamus), the mesencephalon (midbrain, including the red nucleus), and the rhombencephalon, yielding metencephalon (pons and cerebellum) and myelencephalon (medulla).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 939.

The correct answer is: Diencephalon

Question **40**

Not answered

Marked out of 1.00

Flag question

Which of the following is correct with respect to blood brain barrier?

Select one:

- Lipophilic substances penetrate the barrier better
- Basal forebrain lacks the barrier
- Endothelial cells form gap junctions
- Microglial foot processes form the blood brain barrier
- Inflammation can strengthen blood brain barrier

Lipid soluble molecules, such as ethanol and caffeine can penetrate through the barrier relatively easily via the lipid membranes of the cells. Brain endothelial cells are joined by tight junctions of high electrical resistance providing an effective barrier against molecules.

The correct answer is: Lipophilic substances penetrate the barrier better

Question **41**

Not answered

Marked out of 1.00

Flag question

A patient presents with rapid onset personality changes and memory difficulties. On examination, he has homonymous superior quadrantanopsia. Where is the lesion likely to be?

Select one:

- Optic chiasma
- Lateral geniculate body
- Occipital lobe
- Temporal lobe
- Retina

Explanation:

The optic radiations sweep from the lateral geniculate body to the occipital lobe's visual cortex. Some of the fibres sweep anteriorly from the lateral geniculate body into the anterior temporal lobe before heading posteriorly; this detour is known as Meyer's loop. These fibres convey visual information from the contralateral superior quadrant. If a large lesion or a surgical resection of the left temporal lobe damages Meyer's loop, it produces a contralateral

homonymous superior quadrantanopsia.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 280.

The correct answer is: Temporal lobe

Question **42**

Not answered

Marked out of 1.00

Flag question

The part of hypothalamus that acts as a satiety centre is the

Select one:

- Supraoptic nucleus
- Ventromedial hypothalamus
- Paraventricular hypothalamus
- Suprachiasmatic nucleus
- Lateral hypothalamus

Ventromedial hypothalamus acts as a satiety centre; lateral hypothalamus as a feeding centre. In animals with a lesion of the ventromedial hypothalamus abnormal feeding behaviours including hyperphagia that leads to obesity are noted.

The correct answer is: Ventromedial hypothalamus

Question **43**

Not answered

Marked out of 1.00

Flag question

What proportion (in percentage) of the brain is occupied by prefrontal cortex?

Select one:

- 20
- 70
- 50
- 30
- 10

The prefrontal cortex (PFC) is the anterior part of the frontal lobes of the brain, lying in front of the motor and premotor areas, which occupies nearly 30% of the brain.

The correct answer is: 30

Question **44**

Not answered

Marked out of 1.00

Flag question

Huntington's disease is caused due to degeneration of

Select one:

- Pons
- Prefrontal cortex
- Cerebellum

- Striatum
- Subthalamic nucleus

Explanation:

The basal ganglia, a subcortical group of gray matter nuclei, appear to mediate postural tone. The four functionally distinct ganglia are the striatum, the pallidum, the substantia nigra, and the subthalamic nucleus. Collectively known as the corpus striatum, the caudate and putamen harbour components of both motor and association systems. The caudate shrinks dramatically in Huntington disease. This disorder is characterised by rigidity, on which is gradually superimposed choreiform, or 'dancing', movements.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 929.

The correct answer is: Striatum

Question **45**

Not answered

Marked out of 1.00

Flag question

Which of the following is affected if there is an occlusion of the posterior cerebral artery?

Select one:

- Cingulate gyrus
- Corpus callosum
- Dentate gyrus
- Uncinate gyrus
- Arcuate fasciculus

Explanation:

The hippocampus mediates several higher brain functions, such as learning, memory, and spatial coding. The input region of the hippocampus, the dentate gyrus, plays a critical role in these processes. The arterial vascularisation of the hippocampus is dependent on the collateral branches of the posterior cerebral artery and the anterior choroidal artery, forming the network of superficial hippocampal arteries that in turn lead to deep intrahippocampal arteries.

Ref: Jonas P, Lisman J. Structure, function, and plasticity of hippocampal dentate gyrus microcircuits. Frontiers in neural circuits. 2014 Sep 10;8:107.

Tatu L, Vuillier F. Structure and vascularization of the human hippocampus. The Hippocampus in Clinical Neuroscience. 2014;34:18-25.

The correct answer is: Dentate gyrus

Question **46**

Not answered

Marked out of 1.00

Flag question

The motor cortex occupies

Select one:

- Heteromodal association cortices
- Supplementary motor area
- Frontal eye fields

- Precentral gyrus
- Postcentral gyrus

Explanation:

The motor cortex is in the precentral gyrus (Brodmann area 4). Penfield's ground-breaking work defined a motor homunculus in this area, where a somatotropic map of the motor neurons is found. Cells within the motor strip cause contraction of single muscles. The brain region immediately anterior to the motor strip is called the supplementary motor area (Brodmann area 6). This region contains cells that, when stimulated, can trigger more complex movements by influencing a firing sequence of motor strip cells.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 929.

The correct answer is: Precentral gyrus

Question **47**

Not answered

Marked out of 1.00

Flag question

Obstruction to CSF circulation leads to non-communicating hydrocephalus. The cerebrospinal fluid passes into the subarachnoid space via recesses in which of the following structures?

Select one:

- Lateral ventricle
- First ventricle
- Fourth ventricle
- Third ventricle
- Aqueduct of sylvius

CSF is secreted by the choroid plexus in the lateral, third and fourth ventricles and at a rate of 300 ml/day, which is almost protein free. It passes into the subarachnoid space via recesses in the fourth ventricle-Foramina of Magendie and Foramina of Luschka. Obstruction to its circulation commonly occurs within third or fourth ventricle, leading to non-communicating hydrocephalus. Obstruction to CSF flow in the subarachnoid space leads to communicating or normal pressure hydrocephalus.

Circulation of CSF

- 1. From lateral ventricle to 3rd ventricle- via Interventricular foramina of Monro
- 2. From 3rd to 4th Ventricle- via cerebral aqueduct of Sylvius
- 3. From 4th ventricle to subarachnoid space via Foramen of Magendie (single medial foramen) and Foramen of Luschka (two, lateral foramina)

The correct answer is: Fourth ventricle

Question **48**

Not answered

The hippocampus is supplied by

Marked out of 1.00

Flag question

Select one:

- Posterior choroidal artery
- Posterior cerebral artery
- Posterior inferior cerebellar artery
- Middle cerebral artery
- Anterior cerebral artery

Explanation:

The hippocampus mediates several higher brain functions, such as learning, memory, and spatial coding. The input region of the hippocampus, the dentate gyrus, plays a critical role in these processes. The arterial vascularisation of the hippocampus is dependent on the collateral branches of the posterior cerebral artery and the anterior choroidal artery, forming the network of superficial hippocampal arteries that in turn lead to deep intrahippocampal arteries.

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Tatu L, Vuillier F. Structure and vascularization of the human hippocampus. The Hippocampus in Clinical Neuroscience. 2014;34:18-25.

The correct answer is: Posterior cerebral artery

Question **49**

Not answered

Marked out of 1.00

Flag question

Which of the following enzymes is exclusively seen in astrocytes?

Select one:

- COMT
- MAO-B
- Glutamate dehydrogenase
- Tyrosine hydroxylase
- Acetylcholinesterase

Glutamate dehydrogenase is seen only in astrocytes and not in other glial cells.

The correct answer is: Glutamate dehydrogenase

Question **50**

Not answered

Marked out of 1.00

Flag question

In what percentage of left-handed people is the left hemisphere dominant for language?

Select one:

- 90
- 20
- 16
- 10
- 67

Explanation:

In most people, the hemisphere dominant for language also directs the dominant hand. Ninety percent of the population is right-handed, and 99% of right-handers have left hemispheric dominance for language. Of the 10% who are left-handers, 67% also have left-hemispheric language dominance; the other 33% have either mixed or right-hemispheric language dominance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 931.

The correct answer is: 67

Question **51**

Not answered

Marked out of 1.00

Flag question

Lesions in the subthalamic area are associated with which of the following movement disorders?

Select one:

- Hemiballismus
- Huntington chorea
- Dystonia
- Athetosis
- Parkinson disease

Explanation:

Lesions in the subthalamic nucleus yield ballistic movements, sudden limb jerks of such velocity that they are compared to projectile movement. Hemiballismus is a rare and dramatic movement disorder in which a structural lesion or metabolic damage to the subthalamic nucleus causes involuntary flailing, ballistic movements of the limbs.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 929.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 133.

The correct answer is: Hemiballismus

Question **52**

Not answered

Marked out of 1.00

Flag question

Mirror neurons are found in which part of the brain?

Select one:

- Inferior parietal lobe
- Lateral frontal cortex
- Medial-dorsal frontal cortex
- Medial temporal lobe
- Inferior frontal cortex

Functional magnetic resonance imaging studies reveal activity of the inferior frontal and superior parietal cortices both during the eprformance of an action, and the observation of other person performing the same action. These brain regions supposedly contain mirror neurons, which explain why we imitate activity that we observe (modelling). For example while watching sports like tennis the sports fans may feel so emotionally involved in the game and react almost as if they were part of the game themselves. This behavior is explained by the presence of mirror neurons in our brain.

The correct answer is: Inferior frontal cortex

Question **53**

Not answered

Marked out of 1.00

Flag question

Mr. Y developed normal pressure hydrocephalus following trauma in the basal brain. The obstruction is most likely to have occurred in the;

Select one:

- Foramen of Luschka
- Foramen of Munro
- Cisterna Magna
- Arachnoid granulations
- Foramen of Magendie

The question indicates that there is an obstruction. If there is an obstruction after trauma, this is likely to be at Foramen of Monro. A number of cases will have non-obstructive form, where arachnoid villi are implicated.

Posttraumatic hydrocephalus (PTH) is a frequent and serious complication that follows a traumatic brain injury. PTH may present as normal pressure hydrocephalus (NPH, no increase in intracranial pressure) or as a syndrome of increased intracranial pressure due to either communicating or noncommunicating hydrocephalus. In noncommunicating hydrocephalus (obstructive hydrocephalus), CSF accumulates in the ventricles because of CSF flow blockage. As a result, the ventricles enlarge and the hemispheres expand. The commonest site of block in non-communicating hydrocephalus is foramen of Monroe. Other sites include; Third ventricle, Aqueduct of Sylvius, Fourth ventricle, Foramen of Luschka and Foramen of Magendie. Impaired CSF absorption may cause communicating hydrocephalus. The apparent mechanism is partial occlusion of the arachnoid villi in this case. (Posttraumatic Hydrocephalus - Medscape Reference, http://emedicine.medscape.com/article/326411-overview).

The correct answer is: Foramen of Munro

Question **54**

Not answered

Marked out of 1.00

Flag question

Mrs. X is an 83-year-old lady who has been diagnosed with early stages of Alzheimer disease. The brain region that is likely to show prominent changes is

Select one:

- Prefrontal cortex
- Entorhinal cortex
- Superior temporal gyrus
- Caudate nucleus

- Prefrontal cortex

Explanation:

The entorhinal cortex, an essential component of the medial temporal lobe long-term-memory system, represents the main source of input to the hippocampus and the primary target of hippocampal outputs. The involvement of the entorhinal cortex is relevant for neurodegenerative disorders such as Alzheimer disease (AD), as it is one of the earliest affected brain regions. The typical hallmarks of AD, such as the presence of amyloid protein and neurofibrillary tangles, are seen primarily in the entorhinal cortex in mild AD and 'spread' to the hippocampus and other cortical areas as the disease progresses.

Ref: Criscuolo C et al. Entorhinal Cortex dysfunction can be rescued by inhibition of microglial RAGE in an Alzheimer's disease mouse model. Scientific reports. 2017 Feb 13;7(1):1-5.

The correct answer is: Entorhinal cortex

Question **55**

Not answered

Marked out of 1.00

Flag question

What is the function of an endosome?

Select one:

- Recycling cell membrane
- Producing hydrogen peroxide
- Transcribing information from mRNA
- Checking the synthesised proteins for structural conformity before delivery
- Lipid peroxidation

An endosome is a membrane-bound cellular compartment of the endocytic membrane transport pathway extending from the plasma membrane to the lysosome. Molecules internalized from the plasma membrane are transported via endosomes to lysosomes for degradation. Similarly, molecules can also be recycled back to the plasma membrane.

The correct answer is: Recycling cell membrane

Question **56**

Not answered

Marked out of 1.00

Flag question

Which one of the following is a function of oligodendrocytes?

Select one:

- Metabolic clearance of transmitters
- Gliosis and scar formation
- Structural support of neurons
- Contributing to the blood brain barrier
- Myelination

Explanation:

There are three types of glial cells. The most common are the **astrocytes,** which have

several functions, including nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood-brain barrier. Mature astrocytes also express some genes enriched in cell progenitors, suggesting they can retain proliferative potential. The **oligodendrocytes** in the central nervous system and the **Schwann cells** in the peripheral nervous system wrap their processed around neuronal axons, resulting in myelin sheaths that facilitate the conduction of electrical signals. The third type of glial cells, the **microglia**, which are derived from macrophages, are involved in removing cellular debris following neuronal death. The functions of microglia in the non-diseased brain probably include a role in synaptic maintenance.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Siracusa R et al. Astrocytes: role and functions in brain pathologies. Frontiers in pharmacology. 2019 Sep 27;10:1114.

Sasaki A. Microglia and brain macrophages: an update. Neuropathology. 2017 Oct;37(5):452-64.

The correct answer is: Myelination

Question **57**

Not answered

Marked out of 1.00

Flag question

Foramina of Munro are channels that connect

Select one:

- Lateral and third ventricles
- Third and fourth ventricle
- Two third ventricle
- 4th ventricle and spinal canal
- Two lateral ventricles

In the brain, the interventricular foramina of Monro connect the 2 lateral ventricles with the 3rd ventricle at the midline. These foramen serve as conduits for cerebrospinal fluid (CSF) produced in the lateral ventricles to reach the third ventricle and then the rest of the brain's ventricular system.

The correct answer is: Lateral and third ventricles

Question **58**

Not answered

Marked out of 1.00

Flag question

Nucleus accumbens forms a part of which of the following structures of the brain?

Select one:

- Hypothalamus
- Ventral striatum
- Pineal gland
- Dorsal region of thalamus
- Orbitofrontal cortex

The nucleus accumbens is a collection of neurons and forms the main part of the ventral

striatum, located proximal to the head of the caudate nucleus and the anterior portion of the putamen.

The correct answer is: Ventral striatum

Question **59**

Not answered

Marked out of 1.00

Flag question

Synaptogenesis is at its highest during which of the following phases of life?

Select one:

- Preschool years
- Immediately preceding puberty
- During adolescence
- After 65 years of age
- First 2 years of life

Synaptogenesis occurs very rapidly from the second trimester through the first ten years or so of life. The peak of synaptogenesis occurs within the first two postnatal years.

The correct answer is: First 2 years of life

Question **60**

Not answered

Marked out of 1.00

Flag question

The ventral tegmental area is located in which part of the brain?

Select one:

- Midbrain
- Occipital lobe
- Pons
- Medulla
- Temporal lobe

Explanation:

The **midbrain** ventral tegmental area (VTA) lies medial to the substantia nigra and contains dopaminergic neurons that give rise to the mesocorticolimbic dopamine system. These neurons send ascending projections that innervate limbic structures, such as the nucleus accumbens and amygdala; the mesoaccumbens pathway is a central element in the neural representation of reward.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 955.

The correct answer is: Midbrain

Question **61**

Not answered

Marked out of

Which one among the following structures is a constituent of mesencephalon?

Select one:

1.00

Flag question

- Subthalamus
- Cerebellum
- Red nucleus
- Corpus striatum
- Medullary centre

Explanation:

At 4 weeks, the human brain is divided longitudinally into the prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). These three subdivisions or 'vesicles' divide further into five divisions by 5 weeks, consisting of the prosencephalon, which forms the telencephalon (including cortex, hippocampus, and basal ganglia) and diencephalon (thalamus, subthalamus, hypothalamus), the mesencephalon (midbrain, including the red nucleus), and the rhombencephalon, yielding metencephalon (pons and cerebellum) and myelencephalon (medulla).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 939.

The correct answer is: Red nucleus

Question **62**

Not answered

Marked out of 1.00

Flag question

Which of the following is a purely motor cranial nerve?

Select one:

- Facial nerve
- Oculomotor nerve
- Vagus nerve
- Glossopharyngeal nerve
- Vestibulocochlear nerve

Explanation:

The oculomotor nerves (third cranial nerves) originate in the midbrain and supply the pupil constrictor, eyelid, and adductor and elevator muscles of each eye (medial rectus, inferior oblique, inferior rectus, and superior rectus). Oculomotor nerve impairment, a common condition, thus leads to a distinctive constellation: a dilated pupil, ptosis, and outward deviation (abduction) of the eye. Oculomotor nerve injury also impairs the efferent limb of the light reflex. In addition, it impairs the efferent limb of the accommodation reflex, in which the visual system adjusts the shape of the lens to focus on either near or distant objects.

Ref: Kaufman DM, Geyer HL, Milstein MJ, Rosengard JL. Kaufman's Clinical Neurology for Psychiatrists, 9th Edition. 2023. p. 28.

The correct answer is: Oculomotor nerve

Question **63**

Not answered

Which of the following constitute the most common cells of the layer 4 of the cerebral cortex that receives thalamic input?

Marked out of 1.00

Flag question

Select one:

- Large pyramidal cells
- Stellate cells
- Horizontal cells of cajal
- Glial cells
- Small Pyramidal cells

The cerebral cortex is stratified into six layers. The most common cells are pyramidal cells, which are present in all the layers except layer 1 where their dendrites form a tuft. Layer 1 contains no major neuronal cell bodies but glial cells and dendrites from neurones of deeper layers and the horizontal cells of Cajal. Layers 2 and 3 are composed of small pyramidal cells, whose axons project out of and within the hemispheres. Stellate and fusiform cells lie in layer 4 and provide local connections and receive ascending fibres from the thalamus. Layer 5 consists of large pyramidal cells. The cerebellar cortex is three layered. The molecular layer consisting of basket cell and stellate cell, Purkinje layer consisting of Purkinje cell and a granular layer consisting of granule and Golgi cell.

The correct answer is: Stellate cells

Question **64**

Not answered

Marked out of 1.00

Flag question

Electrical synapses are seen in the

Select one:

- Skin
- Olfactory epithelium
- Bone marrow
- Retina
- Lungs

An electrical synapse is formed at a narrow gap between the pre- and postsynaptic neurons (gap junction) and helps conduct electrical messages across cells. Electrical synapses are abundant both in the retina and cerebral cortex of animals.

The correct answer is: Retina

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